

$$\lim a_n \cdot \lim b_n$$

$$=$$

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claim If $\lim_{n \rightarrow \infty} a_n = A$ and $\lim_{n \rightarrow \infty} b_n = B$ then

$$\lim_{n \rightarrow \infty} a_n b_n = AB$$

proof: Given $\varepsilon > 0$, take $M_A, M_B \in \mathbb{N}$ such that

$$n \geq M_A \Rightarrow |a_n - A| < \frac{\varepsilon}{2|B|+2}$$

$$n \geq M_B \Rightarrow |b_n - B| < \frac{\varepsilon}{2M+2} \quad \text{where } M \text{ is an}$$

upper bound of $|a_n|$. Now if $n \geq M_A + M_B$ then

$$|a_n b_n - AB| = |a_n b_n - a_n B + a_n B - AB|$$

$$\leq |a_n| |b_n - B| + |B| |a_n - A|$$

$$< M \cdot \frac{\varepsilon}{2M+2} + |B| \cdot \frac{\varepsilon}{2|B|+2} < \varepsilon \quad \square$$